EM.411 Systems Engineering

Opportunity Set #4 Tradespace Exploration and Concept Selection

Out Date: Wednesday, October 28, 2026
Due Date: Thursday, November 12, 2026 by 10:00am

Administrative Issues

This is a team assignment. The expectation is that all team members should contribute.

This OS addresses the following learning objectives:

SE05	Tradeoffs	Define and explore the system's tradespace to select amongst competing alternatives and objectives.
SE08	Models	Understand the role of rigorous models and the use of simulation and modeling in the system development process.
SA07	Create	Generate several architectures for new or improved systems using both structured and unstructured approaches.
SA08	Select	Select preferred system architectures from a set of architectures, a list of architectural decisions, or a tradespace of architectures

Overview:

In this OS you will explore the design tradespace for a *surface transportation system* in/around the East Cambridge area and present one or two preferred system concepts to your stakeholders.

New stakeholder input has led to the identification of the following priorities for the surface transportation system:

- **Strategic Service Area** – Given the economic growth and increase in numbers of scientific laboratories, major corporations, and high-density residential buildings, you should explore the operation of the proposed surface transportation system in the East Cambridge area, to potentially include: Cambridge Crossing and Lechmere Station, Central Square, and the MIT campus and Kendall Square area. Be sure to note in your report the defined geographic boundary for the system that you used in your analysis.
- **Sustainability** – The new stakeholder input demands the use of battery electric vehicle (BEV) technology. (Comparison to gas-powered, gas-electric, or other propulsion schemes is not expected.)
- **Autonomy** – The new stakeholder input gives preference to the use of vehicle autonomy. You should explore how varying levels of autonomy impact your design.
    - In addition to evaluating autonomous road vehicles, your stakeholders would like you to explore the potential benefits of *autonomous bicycles*[1] that can benefit from the expanding

[1] https://www.media.mit.edu/projects/AutonomousBicycleProject/overview/

bicycle transportation infrastructure in the area. Autonomous bicycles will be power-assisted and are anticipated to navigate to end users to help with first-mile/last-mile transportation. The bicycle does not operate autonomously when a rider is present.

- **Utility Functions** – Related research has provided single-attribute utility (SAU) mappings which can be used in the creation of SAU functions. (Appendix A)

Use the information in this OS (see the system requirements in Table 1 below and Appendices A, B, C, and D) to build your models for the analysis. You may use the modeling software tool of your choice - the tool choice has no influence on grading. We will be looking for clear communication as if to a technically-sophisticated stakeholder, including clarity of insights, well-reasoned conclusions, and good documentation of the model. Please note that you must provide a copy of your model (XLS/XLSX for Excel or source code for MATLAB, Python, etc.) as a separate submission on Canvas. Instructions on how to upload your model are on Page 7.

Table 1: Initial set of system requirements for East Cambridge/Central/MIT/Kendall Square autonomous surface transportation system

Description	Value	Units
Daily Passenger Volume (threshold)	1,500	Passenger Trips/ day
Peak Passenger Throughput (threshold)	150	Passenger Trips/ hour
Off-peak Passenger Throughput (threshold)	50	Passenger Trips/ hour
Desired maximum waiting time for transportation within the system boundary	5	min
Desired maximum transportation time between any 2 locations within the system boundary	7	min
Desired maximum waiting time for transportation outside of the system boundary	20	min

Other Requirements:

- System shall be at least as safe as alternative transportation options
  - NOTE: The city of Cambridge is responsible for road safety of bicycles.
- System should be scalable to meet emergent demand

Constraints:

- Minimum autonomy level[2] within the defined geographic boundary of your proposed system:
  - For road vehicle travel using dedicated lanes: SAE J3016 Level 3
  - Autonomy is built into the e-bike option with a fixed cost for Level 4 autonomy when traveling "empty" to riders for pickup.
- Minimum autonomy level for travel outside the defined geographic boundary of your proposed system (along pre-defined routes or limited-access lanes):
  - SAE J3016 Level 4
- Minimum autonomy level for travel outside the defined geographic boundary of your proposed system (open urban environments):
  - SAE J3016 Level 5

Task 1: Model stakeholder utility (10%)

Single-attribute utility for each of the system measures of performance reflected in the system requirements, as well as the relative importance assigned to those system measures of performance, can be found in *Appendix A*.

Create a multi-attribute utility (MAU) function using the utility data provided in *Appendix A*. Use the MAU function to calculate multi-attribute utilities for these levels of performance seen in these three different architecture cases:

[2] See "SAE J3016 Levels of Driving Automation". https://brx-content.fullsight.org/site/binaries/content/assets/sae-org/content/news/blog/sae-j3016-visual-chart_5.3.21.pdf

Surface transportation system performance	Case A	Case B	Case C
Passenger Volume	1000	2000	750
Peak Passenger Throughput	75	100	75
Average wait time, at MIT/Kendall node	8	12	6
Availability	0.7	0.6	0.8

Report your results in a table. Also, reflect on how the stakeholder priorities found in Appendix A align with the measures of effectiveness (MoEs) your team identified in OS3A and operational objectives your team members identified in OS3B. Would you change the weights assigned to each of the measures of performance in Appendix A based on those insights? If so, to what new values?

Submit your response in **1 PowerPoint slide**, with backup data and formulas (in Excel or .pdf format) included in your model submission (see page 7).

We are looking for: Whether the team has identified an approach discussed in the lectures and applied that approach correctly; Equations defined; Brief discussion of the MAUs for each case and your interpretation of the values; Clear and concise documentation; Discussion of how the SAUs were validated with the given data in Appendix A, how the results were combined into a MAU function, and how the MAU function was validated to calculate reasonable values (reflects preferences); Brief discussion of how accurately you think the stakeholder priorities in Appendix A reflect the overall objectives for the system.

Task 2: Model the technical system (25%)

Create a system model, building on the information provided in the appendices (the model to include possibly both autonomous road vehicles and autonomous bicycles as options). Test your model to verify that it doesn't break down across the range of the system attributes that you will be modeling.

Appendix B contains simplified equations to model an EV (if you use different equations or alter these, please include reasoning).
Appendix C and *Appendix D* contain datasets of components to be used in the design vector[3].

Evaluate different fleet architectures for the *surface transportation system* (possibly including different fleet sizes and compositions). It is acceptable to propose a range (min/max) of potential vehicles in the fleet and a size of the increment between the size groupings that you choose to model (e.g., in increments of 1, 5, 10 vehicles, and so forth).

Develop a system performance model to verify performance against requirements. Draw on the fundamentals of transportation system design (see Appendix B). As you develop your model, you will need to select a primary operating scenario (e.g., based on one of your reports from OS3 — briefly state what the operating scenario is and why you selected it). You should ensure that your system model meets the demands of this operating scenario.

[3] *Design vector*: the small set of architectural decision options which together yields unique combinations (i.e. concepts, or fundamentally different architectures) when one alternative option is selected for each architectural decision.

Document your system model in your PowerPoint slide(s), including additional supporting information (e.g., assumptions are clear, the equations are defined with consistent units). Prepare write-ups or partial tables as needed to demonstrate to a technically-competent peer or supervisor that your model has been verified and validated and is ready to be used to generate valid architectures. Your documentation should have enough information such that someone could duplicate your results using only your model documentation.

In addition to the model documentation, create a table that describes 5 unique architecture instances (combinations of decision-options of your choice). The table should include this information used and/or produced by your model:

- Fleet composition including road vehicles and/or bicycles.
- Fleet size (number of vehicles).
- Vehicle specifications (the design variables from Appendix C or D used in the specific architecture).
- Intermediate performance variables for each architecture instance:
  - Vehicle speed and range; fleet throughput, average wait time, and availability.
- Vehicle and fleet cost.

Submit your response in **3 PowerPoint slides.** At least one slide should be devoted to model documentation. In addition, submit your complete model data and formulas (in Excel or .pdf format) as part of your model submission (see page 7).

We are looking for: All parts of the question are answered; Clear and concise documentation of the model and assumptions, including references to any data sources; Explanation of the method used to verify and validate the model; Units are consistent; The model must produce all specified information (i.e. fleet composition and size, vehicle specifications, intermediate performance variables and costs). Clear reference and validation with the chosen primary operating scenario.

Task 3: Explore the tradespace (20%)

Once you are satisfied that the MAU function and the system models operate properly, combine them and run through the range of the design parameters that represent the range of realistic architecture candidates.

Before running the model, select a reference architecture that you think would be a "good architecture" (based on specific set of design variable choices that are within the range to be explored by the model). Explain briefly why you think that specific choice of architecture variables represents a good set of design choices.

Run the model for the same primary operating scenario you selected in task 2. Explore all the architecture combinations possible in the system model. Create a tradespace plot from the data points your model generated, displaying them along the axes of utility and cost. Clearly identify the Pareto frontier (approximate location is fine) and the utopia point on the tradespace plot.

Identify three distinct architectures along the Pareto frontier that you think represent interesting design points and include a sentence arguing why each architecture is interesting (hint: they will probably be more interesting if they are different from one another based on specific architecture decision choices that create unique functions/capabilities). List those in a table with the same data structure and format as the table in Task 2. Also locate from the model output the same data for the reference architecture

you chose earlier in this task and include it in the table. Indicate the reference architecture using a distinctive marker on the tradespace plot.

As you document and report your findings, you might want to consider these questions: Are there a common set of attributes for those architectures along the Pareto frontier? Is your reference architecture on the Pareto frontier or is it dominated by other architectures? Explain why you think your reference architecture was or was not on the Pareto frontier. Submit your response in **2-3 PowerPoint slides**, referencing your complete model data and formulas in Excel or .pdf format.

We are looking for: All parts of the question are answered; Clear explanations of the assumptions, quality of insights drawn from the exploration and depth of the analysis, including discussion of relevant "what ifs"; Correct use of terminology; Decisions are clearly documented; Graphs are readable and interpretable; The tradespace should have at least eight points (i.e., alternative architectures) on it, if not more; The tradespace is correctly labeled and annotated; Skillful tradespace visualization will be given style points (visualizations and names for concepts, color coding different architectural decisions, etc.). "A picture is worth 1000 words" (when done correctly).

Task 4: Compare architecture performance in a different operating scenario (i.e., system sensitivity to changes) (15%):

Identify another operating scenario for the surface transportation system. Be sure to **not** create an operating scenario that is just a simple variation of the one you previously selected for Task 3. Configure the system model for this operating scenario and run it again. Create a new tradespace graph. Identify the Pareto frontier on this new graph.

Locate the three architectures you reported in the table in Task 3 on the new tradespace, as well as your reference architecture (i.e., put points on the graph to indicate each architecture). Are they still at (or near) the Pareto frontier for this new operating scenario? How has performance for these architecture instances changed from one operating scenario to the other? Is there a new architecture that is better suited to this operating scenario than to the previous one? Explain any new insights you gained from this comparison. Submit your response in **2-3 PowerPoint slides**.

We are looking for: All parts of the question are answered; The quality of your insights; Decisions are clearly documented; As with Task 3, all graphs are readable and interpretable.

Task 5: Recommend a preferred autonomous surface transportation system concept (20%):

Based on the two analyses that you have done, what system concept would you recommend for an autonomous solution for the surface transportation system? Explain how you arrived at this decision. Your recommendation should include a description of the concept, the two operating scenarios, a summary of its performance along the primary attributes prioritized by the stakeholders, and the vehicle and system costs. What architectures (combinations of design variables) would you feel comfortable eliminating from further consideration at this stage? Does the concept that you found to be best through your analysis seem like a realistic solution for surface transportation in the area? Why or why not? Submit your response in **1-2 PowerPoint** slides.

We are looking for: All parts of the question are answered; The data are consistent with those presented in Tasks 3 and 4; The quality and depth of your analysis and reflection; The recommendation is supported by data derived from your model.

Task 6: Discuss the extensibility or expandability of your selected concept to future conditions (10%): You just recommended a preferred system concept which may be in operation for years to come. Optimistically, your recommendation ensures timely and sustained system performance over the full system life cycle. Write a half-page memo reflecting on the life cycle of the system you recommended. As you prepare your memo, you might want to consider these questions: How long will it take to reach desired capacity? Assuming residential and commercial density will change over time along the routes you've specified, how well would your system concept perform as demand changes? What technological changes could pose a threat to or opportunity for your system and is the success of your system dependent on any assumptions? Are some vehicle types more easily upgradeable? How would you try to answer these questions with some degree of certainty? **0.5 PowerPoint** slides.

We are looking for: Answers are reasoned based on your latest understanding of surface transportation systems; Answers account for insights from modeling for preceding 'tasks' where applicable.

Note: Please complete this OS in your Fall 2026 – H2 team. One team member should submit a single response as a team on Canvas, listing the names of all team members who contributed on the cover slide. **This is your main submission containing your responses to the tasks.** Also upload your spreadsheet(s) or supporting file(s) with just your model, data, and formulas in one ZIP file as a separate submission. **Do NOT place your text responses or diagrams in this ZIP, as they will not be graded.** Remember to guide us in your main submission to your additional documents.

Note: We value conciseness; "more" is not always better.

Note: We value your insights and reflections on the formulation, results, and analysis of your model; the model itself does not need to be "production quality".

Note: Please adhere to the slide limits and submit only a single PDF document onto Canvas.

Appendix A – Stakeholder utilities for the MIT/Kendall Square autonomous surface transportation system

Passenger Volume (passenger trips/day)

Passenger trips/day	Utility
0	0
500	0.2
1000	0.4
1500	0.8
2000	1.0

Average wait time, at MIT/Kendall Square node (minutes)

Minutes average wait time	Utility
0	1.0
5	0.95
10	0.75
15	0.4
20	0.2
30	0

Peak Passenger Throughput (passenger trips/hour)

Passenger trips/hour	Utility
0	0
50	0.2
100	0.5
150	0.9
200	1.0

Availability (dml[4])

Availability	Utility
0	0
0.2	0.2
0.4	0.4
0.6	0.6
0.8	0.8
1.0	1.0

Stakeholder priorities

Measure of Performance	Weight
Passenger Volume	0.15
Peak Passenger Throughput	0.25
Average wait time, at MIT/Kendall Square node	0.35
Availability	0.25

[4] Dimensionless

Appendix B - Simplified EV design

Use these equations to build your model.

NOTE: These equations are given as guidelines and as a potential starting point. You are free to adjust as you see fit, but please provide a short justification for doing so.

- Average passenger weight [kg] = 100
- Total Vehicle Cost [dollars] = Chassis Cost + Battery Cost + Charger Cost + Motor and inverter Cost + Autonomy system cost
- Total Fleet Cost [dollars] = Vehicle Cost*Number of vehicles (calculated for each vehicle type and then summed if a mixed fleet is proposed)
- Total Vehicle Weight [kg] = Chassis Weight + Battery Weight + Charger Weight + Motor and inverter Weight + Passengers Weight + Autonomy system Weight
- Battery charge time [h] = Battery Capacity [kWh] / Charger Power [kW]
- Power Consumption [Wh/km] = Nominal Power Consumption $_{\text{Chassis}}$ [Wh/km] + 0.1*(Total Vehicle Weight [kg] - Chassis Weight [kg]) + Added Power Consumption $_{\text{Autonomous System}}$ [Wh/km]
- Range [km] = Battery Capacity [Wh] / Power Consumption [Wh/km]
- Average Speed [km/h] = 700 * Motor Power [kW] / Total Weight [kg]

  NOTE If the calculated average speed is greater than the posted speed limit, then use the posted speed limit. Teams must decide on a speed limit based on their operating scenario.
- Dwell time [s] = 60 (time for passengers get out and get in)
- Expected average load factor/trip = 0.75
- Up-time [h] = Range [km] / Average Speed [km/h]
- Down-time [h] = Battery charge time [h] + 0.25
- Availability [dml] = Up-time / (Up-time + Down-time)
- Benchmark availability of competing systems: 0.75

Average trip demand originating at MIT/Kendall Square (passenger trips/hour)

Time	Demand
0000	15
0200	5
0400	15
0600	50
0800	150
1000	150
1200	150
1400	100
1600	75
1800	100
2000	50
2200	35

Fundamentals of Transportation

You may find information on these two websites useful when building your model. Some relevant extracts are appended below.

- https://en.wikibooks.org/wiki/Fundamentals_of_Transportation/Network_Design_and_Frequency
- https://en.wikibooks.org/wiki/Fundamentals_of_Transportation/Transit_Operations_and_Capacity

Frequency determination [edit]

There are a number of ways that planners may determine a reasonable frequency of service on a route. The most common methods are:[3]

- **Policy headways**. Many transit agencies will determine specific headways that meet policy goals, usually related to providing a minimum level of service along a route when demand on the route is low. This could include 30-minute or 60-minute headways, for example.
- **Minimum headways**. As discussed in the unit on transit operations, bus and rail systems have minimum headways, based on capacity limitations. In cases when demand on a route is very high, these minimum headways can be used to assign the maximum frequencies on the route.
- **Load management**. In many cases where policy headways or minimum headways do not apply, the typical way of determining frequencies is based on managing load at the *peak load point* along the route. The peak load point is that point along the route that experiences the largest number of passengers per hour. Let X be the highest allowable ratio of demand to supply (volume to capacity), with values between 0 and 1 (with 1 when volume = capacity). Then:

$$\frac{P}{f \cdot N_{car} \cdot C_{car}} \leq X \text{ or, } f \geq \frac{P}{X \cdot N_{car} \cdot C_{car}}$$

with $P =$ volume at peak load point (passengers/hr)
$X =$ maximum allowable volume-to-capacity ratio, $0 \leq X \leq 1$
$N_{car} =$ number of cars in a train ($= 1$ for buses)
$C_{car} =$ number of passengers per car

For a bus route operated with 55-passenger buses, a demand of 360 passengers per hour past the peak load point, and an allowable load factor of X = 0.90, the frequency must be:

$$f \geq \frac{360 \text{ passengers/hr}}{0.90 \cdot 1 \cdot (55 \text{ passengers/bus})} = \underline{7.27 \text{ buses/hr}}$$

This would give the planner options for this route such as 7.5 buses/hr (8-min headways) or 8 buses/hr (7.5-min headways), or even higher frequencies.

Capacity [edit]

Capacity in transit operations is measured as the maximum number of passengers that can be carried past a single point on a fixed route, in a given period of time. The most common measure of capacity is in terms of passengers per hour.

In transit operations, we define the time between vehicles past a given point as the *headway* (or time headway), usually in minutes. The inverse of the headway is called the *frequency*, essentially capturing the number of vehicles per unit time past a certain point on a route, usually measured in vehicles per hour.

$$\text{Frequency (veh/hr)} \quad f = \frac{60 \text{ min/hr}}{h}$$

$$\text{Headway (in min)} \quad h = \frac{60 \text{ min/hr}}{f}$$

As an example, a route with a 10-minute headway has a frequency of 6 vehicles per hour.

With this in mind, the capacity of a transit route is given as the product of the frequency and the maximum number of persons per vehicle:

$$\text{Capacity (passengers/hr)} \quad \text{on the route} \; = C_{route} = f \cdot N_{car} \cdot C_{car}$$

$$\begin{aligned} \text{where:} \quad & f = \text{ frequency on the route (veh/hr)} \\ & N_{car} = \text{number of cars per train } (= 1 \text{ for bus}) \\ & C_{car} = \text{maximum number of persons per car (or bus)} \end{aligned}$$

As examples, a bus route with a frequency of 6 buses/hr and a maximum of 55 persons on each bus has a capacity of 330 passengers per hour. A rail line with a frequency of 15 trains/hr (4 min headways) with 6 cars per train and a maximum of 150 persons per car has a capacity of 13,500 passengers per hour.

Sample problem [edit]



Problem:

A transit agency has an existing 8.5-mile bus route (in one direction) where the average speed *between* stops is 16.9 miles per hour. The route has 22 stops along the route. If the average dwell time at each stop is 18 seconds, what is the average bus speed along the route as a whole?

And, if a 5-minute layover is required at the far end of the route, what is the total round-trip travel time? Assume the time for the inbound trip is identical in time to the outbound trip.



Solution:

Total travel time = (8.5 mi) / (16.9 mi/hr) + (22 stops)*(18 sec/stop)*(1 hr / 3600 sec) = 0.613 hr = 37 minutes (rounded up to the nearest minute)

The average speed = (8.5 mi / 0.613 hr) = 13.9 mph.

The total round-trip time = 37 min + 5 min layover + 37 min = 79 min

Appendix C - Dataset for Road Vehicle

Note: data provided assumes mature technology in volume production

Battery Pack

Battery Pack	Capacity [kWh]	Cost [$1000]	Weight [kg]
P1	50	6	110
P2	100	11	220
P3	150	15	340
P4	190	19	450
P5	250	25	570
P6	310	30	680
P7	600	57	1400

Constraints:

- *The battery pack weight shall be no greater than ⅓ of the chassis weight (this is a proxy for limited space availability).*
- *Only one battery pack may be used in a given architecture (no multiple-battery configurations). Each vehicle carries exactly one battery.*

Chassis:

Chassis	Pax	Weight chassis [kg]	Chassis Cost [$1000]	Nominal Power consumption [Wh/km]
C1	2	1350	12	140
C2	4	1600	17	135
C3	6	1800	21	145
C4	8	2000	29	150
C5	10	2200	31	160
C6	16	2500	33	165
C7	20	4000	38	180
C8	30	7000	47	210

Battery charger:

Charger	Power [kW]	Cost on vehicle [$1000]	Weight on vehicle [kg]
G1	10	1	1
G2	20	2.5	1.8
G3	60	7	5

Motor and inverter module:

Motor + Inverter	Weight [kg]	Power [kW]	Cost [$]
M1	35	50	4200
M2	80	100	9800
M3	110	210	13650
M4	200	350	20600

Autonomous system:

SAE Autonomy Level	Autonomy Option	Weight [kg]	Added Power Consumption [Wh/km]	Cost [$1000]
3	A3	30	1.5	15
4	A4	60	2.5	35
5	A5	120	5.0	60

Appendix D – Dataset for Electric Bike

Bike Battery Pack

Battery Pack	Capacity [kWh]	Cost [$1000]	Weight [kg]
E1	0.5	0.6	5
E2	1.5	1.5	11
E3	3	2.6	17

Constraints:

- *The battery pack weight shall be no greater than ½ of the bike frame weight (this is a proxy for limited space availability).*
- *Only one battery pack may be used in a given architecture (no multiple-battery configurations).*

Bike Frame:

Frame#	Pax	Weight [kg]	Frame Cost [$1000]*	Nominal Power consumption [Wh/km]**
B1	1	20	2	30
B2	1	17	3	25
B3	2	35	3.5	40

* NOTE Frame cost and weight includes autonomy Level 4 for the e-bike.

** NOTE As a simplifying assumption, use this spec as the power consumption while autonomously delivering itself (to passengers, charging location, and staging areas, as applicable) as well as the power consumption during rider assist.

Battery Charger:

Charger	Power [kW]	Cost on vehicle [$1000]	Weight on vehicle [kg]
G1	0.2	0.3	0.5
G2	0.6	0.5	1.2

Bike Motor and Inverter Module:

Motor + Inverter	Weight [kg]	Power [kW]	Cost [$]
K1	5	0.35	300
K2	4	0.5	400
K3	7	1.5	600

Appendix E – Deliverable Checklist

This is provided as a reference to help track deliverables. Please double-check the assignment and rubric to ensure your team has addressed all requested information.

Question	Slides	Deliverable	Check
1	1	SAU for each MOP	
		MAU for each case	
		SAU and MAU discussion	
2	3	System model (Excel file, Python, MATLAB, etc.)	
		Model documentation (at least 1 slide)	
		Key model equations	
		Validation approach discussion	
		Architecture table (5 unique architectures)	
3	2-3	Reference architecture definition	
		Tradespace diagram	
		Table with 3 Pareto optimal designs	
		Tradespace analysis and discussion	
4	2-3	Alternate scenario tradespace diagram	
		New recommended architecture	
		Alternate scenario discuission	
5	1-2	Recommendation	
6	0.5	Memo	